EC303 — Signals and System

Module 2: Linear Shift-Invariant (LSI) Systems & Convolution

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Authoritative Syllabus-Aligned Notes

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1. Linear Shift-Invariant (LSI) Systems Overview

Definition: Linear Shift-Invariant (LSI / LTI) System

A system $\mathcal{T}$ is called a **Linear Shift-Invariant (LSI)** or **Linear Time-Invariant (LTI)** system if it satisfies both the principle of **Linearity** (superposition: additivity and homogeneity) and the condition of **Shift-Invariance** (time-invariance).
Formally, for arbitrary scalar multipliers $a, b \in \mathbb{C}$ and arbitrary inputs $x_1(t), x_2(t)$:

1. **Superposition (Linearity):** $\mathcal{T}\{ax_1(t) + bx_2(t)\} = a\mathcal{T}\{x_1(t)\} + b\mathcal{T}\{x_2(t)\} =$

$ay_1(t) + by_2(t)$.

2. **Shift-Invariance (Time-Invariance):** If $\mathcal{T}\{x(t)\} = y(t)$, then $\mathcal{T}\{x(t-t_0)\} = y(t-t_0)$ for any arbitrary shift $t_0 \in \mathbb{R}$ (or $n_0 \in \mathbb{Z}$ in discrete-time).

1.1 Representation of Signals Through Shifted Impulses

Any continuous-time signal $x(t)$ can be viewed as a continuum of weighted, shifted Dirac delta impulses via the sifting property:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \tag{1}$$

Similarly, any discrete-time signal $x[n]$ is exactly decomposed into a linear combination of weighted, shifted Kronecker deltas:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \tag{2}$$

This decomposition forms the conceptual bedrock of LTI system analysis: because the system is linear and time-invariant, knowledge of its response to a single unit impulse ($\delta(t)$ or $\delta[n]$) is sufficient to completely determine its response to *any arbitrary* input signal.

2. Impulse Response and Step Response

2.1 The Impulse Response $h(t)$ and $h[n]$

Definition: Impulse Response

The **Impulse Response** $h(t)$ (in CT) or $h[n]$ (in DT) is the output of an LSI system when the input is a unit impulse applied at the origin under initial rest conditions (zero energy stored in the system):

$$h(t) \triangleq \mathcal{T}\{\delta(t)\} \quad \text{(Continuous-Time)} \tag{3}$$
$$h[n] \triangleq \mathcal{T}\{\delta[n]\} \quad \text{(Discrete-Time)} \tag{4}$$

Physical Significance:

- It acts as the "fingerprint" or signature of an LSI system. Every structural attribute—memory, causality, stability, invertibility, and frequency selectivity—is entirely encoded in $h(t)$ or $h[n]$.
- It characterizes the transient relaxation dynamics of the physical system following an instantaneous excitation.

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2.2 The Step Response $s(t)$ and $s[n]$

Definition: Step Response

The **Step Response** $s(t)$ (in CT) or $s[n]$ (in DT) is the system output when excited by a unit step function $u(t)$ or $u[n]$ under zero initial state:

$$s(t) \triangleq \mathcal{T}\{u(t)\}, \qquad s[n] \triangleq \mathcal{T}\{u[n]\} \tag{5}$$

2.3 Relationship Between Impulse Response and Step Response

Since the unit step is the integral (running sum) of the unit impulse:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau, \qquad u[n] = \sum_{k=-\infty}^{n} \delta[k] \tag{6}$$

By linearity and time-invariance, the step response is the running integral (or running sum) of the impulse response:

Fundamental Inter-relationships: Impulse $\leftrightarrow$ Step Response

Continuous-Time (CT):

$$s(t) = \int_{-\infty}^{t} h(\tau)d\tau \iff h(t) = \frac{ds(t)}{dt} \tag{7}$$

Discrete-Time (DT):

$$s[n] = \sum_{k=-\infty}^{n} h[k] \iff h[n] = s[n] - s[n-1] = \nabla s[n] \tag{8}$$

where ∇ denotes the backward difference operator.

Derivation Example: Step Response of First-Order RC Circuit

Consider a simple low-pass RC series circuit where input $x(t) = v_{in}(t)$ and output $y(t) = v_C(t)$. The differential equation is $RC\frac{dy(t)}{dt} + y(t) = x(t)$.

1. The impulse response is obtained by setting $x(t) = \delta(t)$:

$$h(t) = \frac{1}{RC}e^{-t/RC}u(t)$$

2. The step response $s(t)$ is calculated via running integration for $t \geq 0$:

$$s(t) = \int_{0}^{t} \frac{1}{RC}e^{-\tau/RC}d\tau = \left[-e^{-\tau/RC}\right]_0^t = \left(1 - e^{-t/RC}\right)u(t)$$

3. Differentiating $s(t)$ recovers $h(t)$:

$$\frac{ds(t)}{dt} = \frac{1}{RC}e^{-t/RC}u(t) + (1-e^{-t/RC})\delta(t) = \frac{1}{RC}e^{-t/RC}u(t) + 0 = h(t) \quad (\text{since } 1-e^0 = 0).$$

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3. The Convolution Operation

3.1 Derivation of the Convolution Integral (CT)

Let a continuous-time LSI system $\mathcal{T}$ have impulse response $h(t) = \mathcal{T}\{\delta(t)\}$.

1. Express the input $x(t)$ using the sifting property:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \tag{9}$$

2. Apply the transformation $\mathcal{T}$ to both sides:

$$y(t) = \mathcal{T}\{x(t)\} = \mathcal{T}\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau\right\} \tag{10}$$

3. By linearity (additivity over continuum and homogeneity for scalar $x(\tau)$):

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\mathcal{T}\{\delta(t-\tau)\}d\tau \tag{11}$$

4. By shift-invariance, $\mathcal{T}\{\delta(t-\tau)\} = h(t-\tau)$:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \triangleq x(t) * h(t) \tag{12}$$

3.2 Derivation of the Convolution Sum (DT)

Similarly, for a discrete-time LSI system where $h[n] = \mathcal{T}\{\delta[n]\}$:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \tag{13}$$

$$y[n] = \mathcal{T}\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k]\mathcal{T}\{\delta[n-k]\} = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \triangleq x[n] * h[n] \tag{14}$$

Core Formulation: Convolution Integral and Sum

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau \tag{15}$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \tag{16}$$

3.3 Graphical Interpretation of Convolution

Evaluating convolution graphically consists of four rigorous sequential steps:

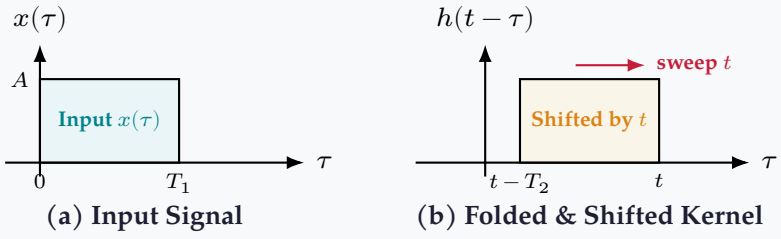
1. **Change of Variable:** Plot $x(\tau)$ and $h(\tau)$ as functions of the dummy variable τ.
2. **Folding (Time Reversal):** Flip $h(\tau)$ about the vertical axis to obtain $h(-\tau)$.

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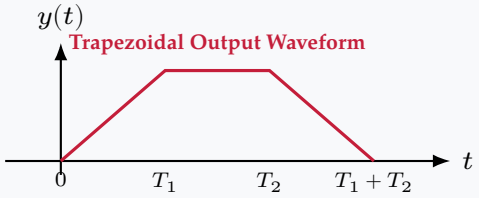
3. **Shifting (Time Delay):** Shift $h(-\tau)$ by time t to obtain $h(t-\tau)$ (right-shift for $t>0$, left-shift for $t<0$).

4. **Multiplication and Integration/Summation:** Multiply $x(\tau)$ and $h(t-\tau)$ pointwise, and evaluate the area under the product curve over the overlapping support interval.

TikZ Illustration: Graphical Convolution Mechanism



(a) Input Signal (b) Folded & Shifted Kernel



(c) Resulting Convolution $y(t) = x(t) * h(t)$

3.4 Properties of Convolution

Property	Continuous-Time Formulation	Discrete-Time Formulation
Commutative	$x(t) * h(t) = h(t) * x(t)$	$x[n] * h[n] = h[n] * x[n]$
Associative	$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$	$[x[n] * h_1[n]] * h_2[n] = x[n] * [h_1[n] * h_2[n]]$
Distributive	$x(t) * [h_1(t) + h_2(t)] = x * h_1 + x * h_2$	$x[n] * [h_1[n] + h_2[n]] = x * h_1 + x * h_2$
Shift Property	$x(t-t_1) * h(t-t_2) = y(t-t_1-t_2)$	$x[n-n_1] * h[n-n_2] = y[n-n_1-n_2]$
Identity Element	$x(t) * \delta(t) = x(t)$	$x[n] * \delta[n] = x[n]$
Derivative Property	$\frac{d}{dt}[x(t) * h(t)] = \frac{dx(t)}{dt} * h(t) = x(t) * \frac{dh(t)}{dt}$	$\nabla[x[n] * h[n]] = \nabla x[n] * h[n] = x[n] * \nabla h[n]$
Duration Property	If lengths are L_1 and L_2, duration of $y(t)$ is $L_1 + L_2$.	Number of non-zero samples is $N_1 + N_2 - 1$.

3.5 Interconnection of LSI Systems

- **Cascade (Series) Connection:** Equivalent impulse response is the convolution of individual impulse responses:

$$h_{eq}(t) = h_1(t) * h_2(t) = h_2(t) * h_1(t) \tag{17}$$

By the commutative property, the order of cascaded LSI subsystems can be interchanged without altering the overall response.

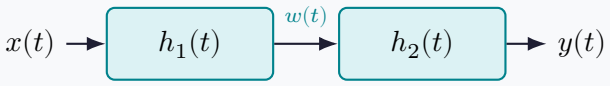
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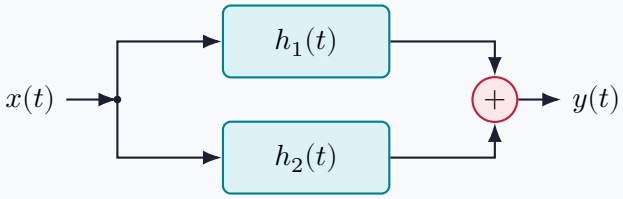
- **Parallel Connection:** Equivalent impulse response is the arithmetic sum of individual responses:

$$h_{eq}(t) = h_1(t) + h_2(t) \tag{18}$$

TikZ Block Diagrams: Cascade and Parallel LSI Interconnections

Cascade Configuration: $h_{eq}(t) = h_1(t) * h_2(t)$





Parallel Configuration: $h_{eq}(t) = h_1(t) + h_2(t)$

4. Causality and Stability of LSI Systems

4.1 Characterization of Causality

A system is causal if the output at any instant t_0 depends only on the input values for $t \leq t_0$.

Causality Criterion for LSI Systems

A continuous-time (or discrete-time) LSI system is **causal** if and only if its impulse response is zero for all negative time:

$$\text{CT:} \quad h(t) = 0 \quad \forall t < 0 \tag{19}$$

$$\text{DT:} \quad h[n] = 0 \quad \forall n < 0 \tag{20}$$

Proof of Causality Condition: The convolution integral gives:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{0} h(\tau)x(t-\tau)d\tau + \int_{0}^{\infty} h(\tau)x(t-\tau)d\tau \tag{21}$$

For $\tau < 0$, the term $t-\tau > t$, representing future input values. Thus, for $y(t)$ to be completely independent of future inputs $x(t-\tau)$ $(\tau < 0)$, we must have:

$$h(\tau) = 0 \quad \text{for all } \tau < 0 \tag{22}$$

Under causality, the convolution limits simplify to:

$$y(t) = \int_{0}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)h(t-\tau)d\tau \tag{23}$$

If the input $x(t)$ is also causal $(x(t) = 0$ for $t < 0)$:

$$y(t) = \int_{0}^{t} x(\tau)h(t-\tau)d\tau \quad (t \geq 0) \tag{24}$$

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4.2 Characterization of BIBO Stability

A system is Bounded-Input Bounded-Output (BIBO) stable if every bounded input $|x(t)| \leq M_x < \infty$ produces a bounded output $|y(t)| \leq M_y < \infty$.

BIBO Stability Criterion for LSI Systems

An LSI system is **BIBO stable** if and only if its impulse response is **absolutely integrable** (in CT) or **absolutely summable** (in DT):

$$\text{CT:} \quad \int_{-\infty}^{\infty} |h(t)|dt < \infty \tag{25}$$

$$\text{DT:} \quad \sum_{n=-\infty}^{\infty} |h[n]| < \infty \tag{26}$$

Rigorous Proof of BIBO Stability Criterion (Sufficiency and Necessity)

1. Sufficiency Proof (If $\int |h(t)|dt < \infty \implies$ BIBO Stable): Let $|x(t)| \leq M_x < \infty$. Taking the magnitude of the output:

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)| \cdot |x(t-\tau)|d\tau$$
$$\leq M_x \int_{-\infty}^{\infty} |h(\tau)|d\tau$$

Since $\int_{-\infty}^{\infty} |h(\tau)|d\tau = K < \infty$, we have $|y(t)| \leq M_x K < \infty$ for all t. Hence, the output is guaranteed bounded.

2. Necessity Proof (If BIBO Stable $\implies \int |h(t)|dt < \infty$): Suppose $\int_{-\infty}^{\infty} |h(t)|dt = \infty$. Choose the bounded input:

$$x(t-\tau) = \text{sgn}[h(\tau)] = \begin{cases} +1, & h(\tau) > 0 \\ -1, & h(\tau) < 0 \end{cases}$$

Clearly $|x(t)| = 1 \leq 1$ is bounded. Then at time $t = 0$:

$$y(0) = \int_{-\infty}^{\infty} h(\tau)x(-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)\text{sgn}[h(\tau)]d\tau = \int_{-\infty}^{\infty} |h(\tau)|d\tau = \infty$$

The output grows unbounded at $t = 0$, violating BIBO stability. Therefore, absolute integrability is both necessary and sufficient.

5. System Representation Through Differential Equations

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5.1 Linear Constant-Coefficient Differential Equations (LCCDE)

A continuous-time lumped parameter LTI system is generally modeled by an N-th order LCCDE relating input $x(t)$ to output $y(t)$:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^{M} b_m \frac{d^m x(t)}{dt^m} \tag{27}$$

where a_k, b_m are real, time-invariant coefficients, and $N \geq M$ for physical realizability.

5.2 Complete Solution: Natural and Forced Responses

The total output consists of two components:

$$y(t) = y_h(t) + y_p(t) = y_{zi}(t) + y_{zs}(t) \tag{28}$$

- **Homogeneous Solution $y_h(t)$ (Natural Response):** Solution when input $x(t) \equiv 0$:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = 0 \implies \sum_{k=0}^{N} a_k \lambda^k = 0 \quad \text{(Characteristic Equation)} \tag{29}$$

  If roots λ_i are distinct: $y_h(t) = \sum_{i=1}^{N} C_i e^{\lambda_i t}$.

- **Particular Solution $y_p(t)$ (Forced Response):** The specific response satisfying the non-zero input excitation $x(t)$.
- **Zero-Input Response (ZIR) $y_{zi}(t)$:** Response produced purely by internal stored energy (initial conditions) with zero external input.
- **Zero-State Response (ZSR) $y_{zs}(t)$:** Response produced purely by the input with all initial conditions set to zero: $y_{zs}(t) = x(t) * h(t)$.

5.3 Condition of Initial Rest

Concept: Initial Rest

An LCCDE system is said to be at **Initial Rest** if the system starts with zero energy storage prior to input application:

$$\text{If } x(t) = 0 \text{ for } t \leq t_0 \implies y(t) = 0 \text{ for } t \leq t_0 \tag{30}$$

Initial rest is the necessary and sufficient condition that guarantees the differential equation system is **Linear**, **Time-Invariant**, and **Causal**.

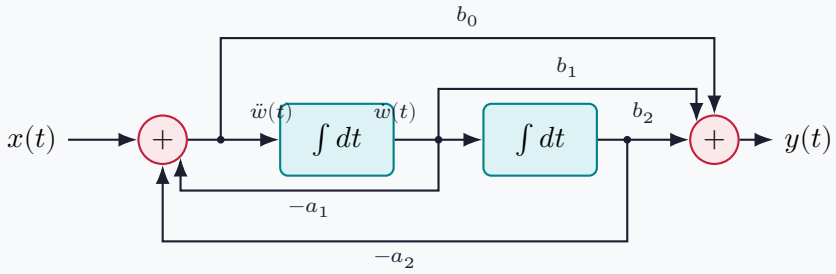
5.4 Block Diagram Realization of LCCDE Systems

LCCDEs can be physically synthesized using three elementary operational building blocks:

1. **Integrators ($\int$):** preferred over differentiators because differentiators amplify high-frequency noise.
2. **Adders / Summers ($\sum$):** compute algebraic additions.
3. **Scalers / Multipliers (a_k, b_m):** amplify signals by constant gains.

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TikZ Schematic: Direct Form II (Canonic) Realization of 2nd-Order LCCDE



6. Behavior with Aperiodic Convergent Inputs

When an aperiodic convergent input signal (such as an exponentially decaying signal $x(t) = e^{-at}u(t)$, $a > 0$) is applied to an LSI system:

- **Energy Preservation:** If both $x(t)$ and $h(t)$ are L_1 and L_2 bounded (energy signals), the output $y(t) = x(t) * h(t)$ is also guaranteed to be an energy signal:

$$E_y = \int_{-\infty}^{\infty} |y(t)|^2 dt < \infty \tag{31}$$

- **Convergence to Zero:** Since $\lim_{t\to\infty} x(t) = 0$, for any stable system with $\int |h(t)|dt < \infty$, the output asymptotically decays to zero:

$$\lim_{t\to\infty} y(t) = \lim_{t\to\infty} \int_0^t x(\tau)h(t-\tau)d\tau = 0 \tag{32}$$

- **Transient Decay Rate:** The overall decay envelope is governed by the slowest pole (dominant time constant) among the system and the input excitation.

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Quick Revision & Examination Compendium

Module 2: Linear Shift-Invariant Systems & Convolution

Key Formulas at a Glance

Concept / Law	Mathematical Equation
Continuous-Time Convolution	$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$
Discrete-Time Convolution	$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$
Step Response from Impulse	$s(t) = \int_{-\infty}^{t} h(\tau)d\tau, \quad s[n] = \sum_{k=-\infty}^{n} h[k]$
Impulse Response from Step	$h(t) = \frac{ds(t)}{dt}, \quad h[n] = s[n] - s[n-1]$
LTI Causality Condition	$h(t) = 0 \quad \forall t < 0, \qquad h[n] = 0 \quad \forall n < 0$
LTI BIBO Stability Condition	$\int_{-\infty}^{\infty} \lvert h(t)\rvert dt < \infty, \qquad \sum_{n=-\infty}^{\infty} \lvert h[n]\rvert < \infty$
Cascade Connection	$h_{eq}(t) = h_1(t) * h_2(t) = h_2(t) * h_1(t)$
Parallel Connection	$h_{eq}(t) = h_1(t) + h_2(t)$
Convolution with Shifted Impulse	$x(t) * \delta(t - t_0) = x(t - t_0)$
LCCDE Zero-State Output	$y_{zs}(t) = x(t) * h(t)$ under initial rest

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One-Line Definitions for Rapid Review

Term	Definitive Technical Meaning
Impulse Response $h(t)$	The output response of an LTI system when excited by a unit impulse under zero initial conditions.
Step Response $s(t)$	The response of an LTI system to a unit step input under initial rest conditions.
Convolution	A mathematical operator combining two signals to express how the shape of one is modified by the other.
Initial Rest	The condition where a system has zero internal energy stored prior to the instant an input is applied.
Causal LSI System	An LTI system whose impulse response is strictly zero for all negative time ($h(t) = 0, t < 0$).
BIBO Stable LSI System	An LTI system whose impulse response is absolutely integrable ($\int \lvert h(t) \rvert dt < \infty$).
Zero-Input Response (ZIR)	The system output generated solely by non-zero initial states in the absence of any input.
Zero-State Response (ZSR)	The system output generated solely by the external input when all initial states are zero.
Canonic Realization	A block diagram realization containing the minimum possible number of delay or integrator elements (N).

Frequently Asked / Viva Questions

Q1. Why does the impulse response $h(t)$ completely characterize an LSI system?

Because any input $x(t)$ can be represented as an integral of weighted, shifted impulses. By linearity (superposition) and time-invariance, the output is the corresponding integral of weighted, shifted impulse responses. Thus, knowing $h(t)$ determines the response to *any* arbitrary input.

Q2. How can you compute the impulse response if the step response is known?

For continuous-time systems, differentiate the step response: $h(t) = \frac{ds(t)}{dt}$. For discrete-time systems, take the first backward difference: $h[n] = s[n] - s[n-1]$.

Q3. Prove that convolution is commutative: $x(t) * h(t) = h(t) * x(t)$.

By definition, $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$. Substituting $v = t - \tau \implies d\tau = -dv$. When $\tau = -\infty, v = +\infty$; when $\tau = +\infty, v = -\infty$. Thus: $\int_{+\infty}^{-\infty} x(t - v)h(v)(-dv) = \int_{-\infty}^{\infty} h(v)x(t-v)dv = h(t) * x(t)$.

Q4. What is the total duration of the convolution of two signals of lengths T_1 and T_2?

In continuous-time, the duration is $T_1 + T_2$. In discrete-time, if signal 1 has N_1

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Frequently Asked / Viva Questions (contd.)

samples and signal 2 has N_2 samples, the convolved output has exactly N_1+N_2-1 samples.

Q5. State the necessary and sufficient condition for an LSI system to be causal.

The impulse response must be zero for all negative time: $h(t) = 0$ for $t < 0$ (or $h[n] = 0$ for $n < 0$).

Q6. Is the system with impulse response $h(t) = e^{2t}u(t)$ stable? Why?

No. Evaluating absolute integrability: $\int_{-\infty}^{\infty} |h(t)|dt = \int_0^{\infty} e^{2t}dt = \left[\frac{e^{2t}}{2}\right]_0^{\infty} = \infty$. Since it is not absolutely integrable, it is BIBO unstable.

Q7. What is the difference between Zero-Input Response (ZIR) and Zero-State Response (ZSR)?

ZIR is driven purely by initial energy/conditions stored in capacitors/inductors with zero external input. ZSR is driven purely by the applied input with zero initial stored energy. Total response is $y(t) = y_{zir}(t) + y_{zsr}(t)$.

Q8. Why does the condition of initial rest ensure that a differential equation represents an LTI system?

Non-zero initial conditions introduce an additive constant independent of the input, violating linearity (homogeneity $y(0) \neq 0$) and causing time-variant behavior. Initial rest enforces zero state prior to excitation, ensuring linearity, time-invariance, and causality.

Q9. Why are integrators preferred over differentiators in analog computer simulation of LCCDE?

Differentiators have transfer function s, which amplifies high-frequency noise and introduces signal instability. Integrators have transfer function $1/s$, which attenuates high frequencies, acting as low-pass filters that provide stable, noise-free computation.

Q10. What is the convolution of a signal $x(t)$ with a unit step $u(t)$?

$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau)u(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)d\tau$. Convolution with a unit step is equivalent to running integration of the signal from $-\infty$ to t.

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Summary Matrix: LSI Characterization Across Domains

Property	Continuous-Time (CT)	Discrete-Time (DT)
Impulse Input	**Dirac Delta** $\delta(t)$	Kronecker Delta $\delta[n]$
Convolution	$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$	$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$
Causality	$h(t) = 0 \quad \forall t < 0$	$h[n] = 0 \quad \forall n < 0$
BIBO Stability	$\int_{-\infty}^{\infty} \lvert h(t)\rvert dt < \infty$	$\sum_{n=-\infty}^{\infty} \lvert h[n]\rvert < \infty$
Step Response s	$s(t) = \int_{-\infty}^{t} h(\tau)d\tau$	$s[n] = \sum_{k=-\infty}^{n} h[k]$
Recovery of h	$h(t) = \frac{ds(t)}{dt}$	$h[n] = s[n] - s[n-1]$
Interconnection	**Cascade:** $h_1 * h_2$**; Parallel:** $h_1 + h_2$	Cascade: $h_1 * h_2$; Parallel: $h_1 + h_2$

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